

Zephyr DM9051 驅動快速指南

核心檔案清單

類型	檔案路徑	說明
驅動	zephyr/drivers/ethernet/eth_dm9051.c	驅動主程式
	zephyr/drivers/ethernet/eth_dm9051_priv.h	暫存器定義與內部結構
建置	zephyr/drivers/ethernet/CMakeLists.txt	建置腳本
Kconfig	zephyr/drivers/ethernet/Kconfig	Kconfig 文檔
	zephyr/drivers/ethernet/Kconfig.dm9051	配置選項
Device Tree	zephyr/dts/bindings/ethernet/davicom,dm9051.yaml	硬體綁定定義
範例	samples/net/dhcpv4_client/overlay_nrf54l15_dm9051.overlay	應用配置範例

Kconfig.dm9051 配置選項定義

CONFIG_ETH_DM9051=y	# 啟用驅動
CONFIG_ETH_DM9051_DEBUG_PRINTS=n	# 封包處理超時
CONFIG_ETH_DM9051_RX_THREAD_STACK_SIZE=800	# RX 執行緒堆疊 (預設)
CONFIG_ETH_DM9051_RX_THREAD_PRIO=2	# RX 執行緒優先權

Device Tree 屬性

屬性	必要	範例	說明
compatible	✓	"davicom,dm9051"	固定值
reg	✓	<0>	SPI CS 編號
spi-max-frequency	✓	<16000000>	SPI 時脈 (8~40MHz)
int-gpios	X	<&gpio0 3 GPIO_ACTIVE_LOW>	中斷腳位 (省略則 Polling)
reset-gpios	X	<&gpio2 0 GPIO_ACTIVE_LOW>	重置腳位 (省略則無硬體重置)
local-mac-	X	[00 60 6E 12 34 56]	MAC 位址 (預設全零)

屬性	必要	範例	說明
address			

快速配置範例

中斷模式 (含硬體重置)

```

&gpio0 { status = "okay"; };
&gpio1 { status = "okay"; };
&gpio2 { status = "okay"; };
&gpiote20 { status = "okay"; };
&gpiote30 { status = "okay"; };

&pinctrl {
    spi21_default: spi21_default {
        group1 {
            psels = <NRF_PSEL(SPIM_SCK, 1, 3)>,
                    <NRF_PSEL(SPIM_MOSI, 1, 1)>,
                    <NRF_PSEL(SPIM_MISO, 1, 0)>;
        };
    };
    spi21_sleep: spi21_sleep {
        group1 {
            psels = <NRF_PSEL(SPIM_SCK, 1, 3)>,
                    <NRF_PSEL(SPIM_MOSI, 1, 1)>,
                    <NRF_PSEL(SPIM_MISO, 1, 0)>;
            low-power-enable;
        };
    };
};

&spi21 {
    status = "okay";
    cs-gpios = <&gpio0 10 GPIO_ACTIVE_LOW>;
    pinctrl-0 = <&spi21_default>;
    pinctrl-1 = <&spi21_sleep>;
    pinctrl-names = "default", "sleep";

    dm9051@0 {
        compatible = "davicom,dm9051";
        reg = <0>;
        spi-max-frequency = <16000000>;
        int-gpios = <&gpio0 3 GPIO_ACTIVE_LOW>;
        reset-gpios = <&gpio2 0 GPIO_ACTIVE_LOW>;
        local-mac-address = [00 60 6E 12 34 56];
    };
};

```

```
};  
};
```

Polling 模式 (無中斷、無重置)

```
&gpio0 { status = "okay"; };  
&gpio1 { status = "okay"; };  
  
&pinctrl {  
    spi21_default: spi21_default {  
        group1 {  
            psels = <NRF_PSEL(SPIM_SCK, 1, 3)>,  
                  <NRF_PSEL(SPIM_MOSI, 1, 1)>,  
                  <NRF_PSEL(SPIM_MISO, 1, 0)>;  
        };  
    };  
};  
  
&spi21 {  
    status = "okay";  
    cs-gpios = <&gpio0 10 GPIO_ACTIVE_LOW>;  
    pinctrl-0 = <&spi21_default>;  
    pinctrl-names = "default";  
  
    dm9051@0 {  
        compatible = "davicom,dm9051";  
        reg = <0>;  
        spi-max-frequency = <8000000>;  
        local-mac-address = [00 00 00 00 00 00];  
    };  
};
```

啟用步驟

1. 在 prj.conf 設定啟用驅動選項

```
CONFIG_ETH_DM9051=y  
CONFIG_SPI=y  
CONFIG_GPIO=y
```

2. 創建 Device Tree Overlay

```
/* 參考上方範例配置 */
```

3. 在 CMakeLists.txt 引用 dm90驅動源碼

```
zephyr_library_sources_ifdef(CONFIG_ETH_DM9051    eth_dm9051.c)
```

4. 編譯並燒錄

```
west build -b nrf54l15dk/nrf54l15/cpuapp  
west flash
```

重要提醒

- **SPI 頻率**：建議從 8MHz 開始測試，穩定後可提升至 16~40MHz
- **MAC 位址**：[00 00 00 00 00 00] 會使用晶片預設值
- **Polling vs 中斷**：移除 `int-gpios` 即自動切換為 Polling 模式
- **RESET Pluse**：添加 `reset-gpios` 即自動加上開機reset訊號波